Teacher:	Ray Joshua Balingkit	**Subject:**	Design & Technology
Week:	12 (Term 1)	**Year:**	10–11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will classify material families and relate working properties to product use. They will develop their understanding through a teacher explanation, a worked example and an evidence-based classroom task. In the second lesson, students will compare paper, polymers, metals and timbers for a defined product component. Their recorded reasoning, practical evidence and final checks will show whether they can apply the ideas to a defined user, product or design problem.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher questioning, labelled diagrams, case comparisons and the completed independent tasks. Check accurate terminology, evidence linked to the user or brief and a reasoned explanation of choices. Review recorded tests or traces alongside conclusions, and use the final check in both lessons to identify the next skill needing practice.

LESSON SLOTS - TUESDAY AND WEDNESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	10 November 2026	By the end of this lesson, students will be able to classify material families and relate working properties to product use.	**H:** Present paper, a polymer item, a metal spoon and a timber offcut or photographs. Ask why appearance alone may not reveal a material's properties or how it can be worked. **ID:** Introduce the book's core material families: metals, papers and boards, polymers, fibres and textiles, and natural and manufactured timbers. Explain physical properties and working properties through stiffness, toughness, hardness, conductivity, formability and ease of joining. Clarify that properties vary between specific materials within a family. **WD:** Compare a paperboard package with a polymer container. Identify the actual material where known, explain folding versus moulding	**Support:** Provide a glossary and a table with family names already entered. Use touch-safe prepared samples and match one property to one use before requiring two linked explanations. **Extension:** Students identify an exception within a family and explain why the exact material grade and product form should be specified before manufacture.

			and connect the choice to the product. Model qualifying an uncertain identification rather than claiming that every similar-looking material behaves alike. **YD:** Students create a family-property-use table from teacher-prepared samples and information cards. They identify each family, record two relevant properties and explain a suitable use. They distinguish an observation they made from a property supplied by the data card. **C:** Ask students to distinguish stiffness from strength and a physical property from a working consideration. Check that examples use specific material evidence rather than treating all metals or polymers as identical.	
2	11 Novembe r 2026	By the end of this lesson, students will be able to compare paper, polymers, metals and timbers for a defined product component.	**H:** Present a desk organiser brief and ask whether the cheapest sheet material necessarily produces the lowest-cost durable product. Invite students to consider making, joining and expected use. **ID:** Explain comparison through a common set of criteria: stiffness, mass, durability, moisture resistance, ease of shaping, joining and cost. Introduce ferrous and non-ferrous metals, thermoforming and thermosetting polymers, and natural and manufactured timbers at recognition level using the book's selected examples. **WD:** Model a comparison of card, a named polymer sheet, aluminium and plywood using supplied evidence. Explain why card may suit a prototype while another material	**Support:** Offer a comparison grid, material cards and highlighted essential criteria. Compare two choices first, then use the same questions for the remaining materials. **Extension:** Students propose a small comparative test with controlled dimensions and loading, explain its limits and identify why a single test cannot establish every property.

			better suits repeated use. Distinguish choosing a model material from specifying a final production material. **YD:** Students complete a four-material comparison for a stationery holder and recommend separate materials for a prototype and a finished product. They explain two decisive criteria, one rejected option and how a manufacturing or joining constraint affects the recommendation. **C:** Ask what changes if the holder must withstand damp conditions or be easily disassembled. Check that the revised selection responds to the changed need and still uses the evidence provided.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel GCSE (9–1) Design and Technology revision notes: 1.8–1.12 Material families and properties
- Teacher computer, display screen and whiteboard markers
- Prepared product scenarios, diagrams and user briefs aligned to the lesson objectives
- Prepared analysis worksheet, comparison grid or test-recording table
- Paper, pencils, rulers, card templates and approved classroom stationery
- Teacher-prepared samples or component cards; pre-cut model parts where required
- Lesson success criteria, observation checklist and labelled evidence folders



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?